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Project Kamp

Internal Project for
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Project Description

Bedriften Consulting Limited is developing a web-based platform to manage the Tennis Tournaments in Ghana. The platform will include a real-time scoring app for umpires, public-facing tournament site, and live video streaming capabilities. The primary goal is to streamline tournament operations, increase fan engagement, and provide real-time updates for players, organizers, and spectators

Scope

Objectives:

- Digitize match scoring via an umpire web app.
- Feed live scores to the main tournament website.
- Display live brackets, player profiles, and match schedules.
- Enable live video streaming of selected matches.
- Provide a central dashboard for organizers to manage tournament logistics.

System Components

1. Umpire Scoring App
    a. Accessible via tablets/mobile phones
    b. Match selection based on assigned umpire
    c. Real-time point-by-point scoring
    d. Auto-calculation of sets and match winner
    e. Syncs to backend and updates live site

2. Tournament Website
    a. Public access
    b. Features:
        i. Home: Tournament info, schedule, and highlights
        ii. Live Scores: Auto-refreshing scores from matches
        iii. Brackets: Visualized match progress (single-elimination)
        iv. Player Profiles: Bios, rankings, past performance
        v. Video Streams: Embedded live matches (YouTube or Vimeo integration)
        vi. Admin Panel (secured): Match assignments, user roles, manual overrides

3. Admin Dashboard
    a. Create/manage players, brackets, and schedules
    b. Assign umpires to matches
    c. Manage court allocations
    d. Real-time monitoring of ongoing matches
    e. Push notifications for next matches

Functional Requirements

For Umpires:

1. Login with unique credentials
2. View assigned match only
3. Score input buttons: “+15” per point, Undo last point, Deuce handling, Tie-Break
4. Automatic game, set, and match status updates
5. Pause/resume options
6. End match & confirm winner

For Spectators:

1. View live scores
2. Watch live streams
3. Browse brackets and results
4. Access player information and match stats

For Admins:

1. Full CRUD for users, matches, and players
2. Match scheduling interface
3. Manual result override in case of tech failure
4. Court and umpire assignment
5. Match event logging

Deliverables

1. Mobile (Android) /Web App for Umpire Scoring
2. Tournament Website
3. Admin Portal

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Timelines

Phase	Activity	Duration (Days)
Phase 1	User Requirement Gathering and Design	1 Day
Phase 2	Website Design and Development	5 Days
Phase 3	Mobile App Development	3 Days
Phase 4	Admin Portal	3 Days
Phase 5	QA and Testing	2 Days
Phase 6	GO-LIVE	1 Day
Phase 7	Post-Implementation Support	30 Days
Total		45 Days

Days mean Calendar Days (Monday – Sunday)

Technical Requirements

Layer / Category	Component	Technology / Service
Presentation Layer	Frontend Web Application	React (TypeScript)
Application Layer	Backend APIs	GoLang (RESTful APIs)
Reporting & Analytics	Reporting Engine	Python
Monitoring & Observability	Monitoring Dashboard	Grafana
Cloud Infrastructure	Hosting Environment	AWS Cloud
Database Layer	Relational Database	AWS RDS (PostgreSQL/MySQL)
Database Layer	Caching	Redis (AWS ElastiCache)
Storage	Object Storage	AWS S3
Authentication & Security	Identity Management	AWS Cognito / OAuth 2.0
	Encryption	TLS 1.2+, AWS KMS
	Web Application Firewall	AWS WAF
	Secrets Management	AWS Secrets Manager
Payment Integration	Payment Gateway APIs	Mobile Money via Fidelity Bank
Messaging & Notifications	Email & SMS Services	MNotify
CI/CD & DevOps	Pipeline Automation	GitHub Actions
Containerization	Application Packaging	Docker
Container Orchestration	Cluster Management	AWS ECS / EKS (Kubernetes)
Logging & Audit	Centralized Logging	AWS CloudWatch / ELK Stack
Compliance & Audit	Audit Logging	AWS CloudTrail
Backup & DR	Automated Backups	AWS Backup
	Disaster Recovery	Multi-Region Replication
Fraud Prevention	Risk Detection	Custom Logic + AWS Fraud Detector
Scalability	Load Balancing	AWS Elastic Load Balancer
Content Delivery	CDN	AWS CloudFront
Non-Functional Requirements	Availability	99.9% Uptime SLA
	Performance	<3s Page Load Time
	Security Model	RBAC + Least Privilege
	Compliance	Data Protection Act (Ghana), PCI-DSS, ISO 27001
	Auditability	Transparent Logging & Reporting

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Thank you

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